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Master Thesis

“Deep Learning for Photomultiplier Crystal Detection Signals”

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Deep Learning for Photomultiplier Crystal Detection Signals

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Abstract—**TODO:** 150–250 palabras. Problema, metodo propuesto, datos empleados y resultados cuantitativos principales. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

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Index Terms—Positron emission tomography, silicon photomultiplier, missing data imputation, deep learning, graph neural networks, silicon photomultiplier failure.

I. INTRODUCTION

Positron Emission Tomography (PET) is a non-invasive molecular imaging modality and one of the central techniques of nuclear medicine, providing an *in vivo* assessment of the metabolic and biochemical state of a patient [1]. Unlike purely anatomical modalities, PET relies on radiotracers: biologically active molecules chemically labelled with positron-emitting radionuclides [2]. Once administered, the tracer distributes according to the physiological pathway targeted by its carrier molecule, so that the measured signal reflects function rather than structure. When an emitted positron annihilates with an electron in the surrounding tissue, two photons of 511 keV — the rest mass energy of the electron — are released in nearly opposite directions. Detecting both of them in coincidence defines a line of response (LOR) along which the annihilation

took place, and the accumulation of millions of LORs allows the three-dimensional distribution of activity to be reconstructed [1]. This detection scheme gives PET a functional sensitivity that few other non-invasive imaging techniques can match [3], at the price of the tracer itself: positron-emitting radionuclides are radioactive by nature, costly to produce, and short-lived, which ties any PET facility to a nearby cyclotron and specific facilities [2].

This functional readout is most valuable where disease alters metabolism before it alters anatomy. In oncology, ^{18}F -FDG PET has become a cornerstone for diagnosis, staging, restaging and the assessment of treatment response [4], [5]. Its role in neurology is arguably even more distinctive, since it reveals pathologies that produce no macroscopic structural abnormality on conventional Magnetic Resonance Imaging (MRI) or Computed Tomography (CT), such as the characteristic hypometabolic and amyloid patterns of Alzheimer's disease [6], among other neurodegenerative and cerebrovascular disorders.

The sensitivity of PET is not matched by its spatial resolution. Clinical scanners typically resolve 4–5 mm full width at half maximum (FWHM) [7], and recent advances push the number towards 3 mm [7], [8], higher than the numbers normally achieved with MRI or CT. This limit decomposes into four contributions, two physical and two belonging to the detector, which behave very differently from one another [9]. The physical ones are the finite distance the positron travels before annihilating, set by the emitting isotope [10], and the small angular deviation from antiparallel emission, caused by the residual momentum of the electron–positron pair. Neither can be engineered away, although the second translates into a positional error that scales with the diameter of the detector ring and is therefore smaller in compact systems. The detector terms are more tractable. In pixelated arrays, the size of the individual element fixes the finest scale at which an interaction can be localised; shrinking it improves resolution, but the inter-crystal reflectors then occupy a growing fraction of the detector volume, and the probability that a photon deposits energy across several elements rises. The decoding error — the uncertainty in determining where within the module the interaction occurred — amounts to roughly one third of the detector width [9]. Unlike positron range and non-collinearity, this term is not a physical floor but a consequence of how the light distribution is read and interpreted, and it is therefore the one most open to improvement, whether through detector geometry, readout

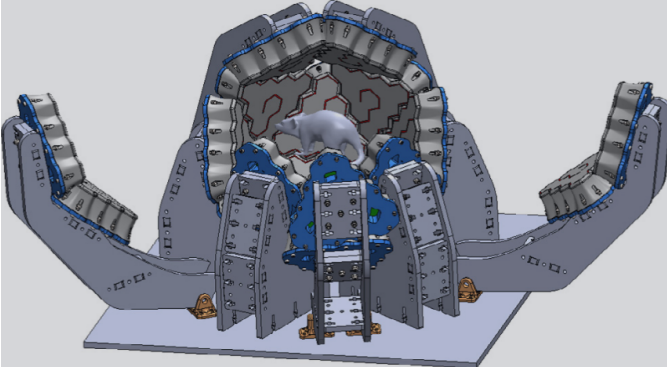


Figure 1: **TEMPORAL** Three-dimensional model of the quasi-spherical, hexagonal-based PET scanner. Reproduced from [12].

electronics or the algorithms that interpret the signal. This is the ground occupied by organ-dedicated scanners. Adapting the detector geometry to a specific anatomical region increases solid-angle coverage, and with it sensitivity and resolution, at a lower manufacturing cost than a general-purpose whole-body system; designs exist for breast, brain, prostate and cardiac imaging [11]. The trade-off is that irregular geometries complicate calibration and data correction, and that small gantry apertures accentuate parallax errors and degrade the uniformity of spatial resolution across the field of view [11]. Much of the design effort in these systems therefore shifts onto the detector itself.

Our group is currently developing a quasi-spherical brain-dedicated PET scanner along these lines [12]. Rather than the cylindrical ring of conventional systems, its detector modules tile a closed surface around the head (Fig. 1), approaching full 4π steradian coverage and thereby collecting a far larger fraction of the emitted coincidences [12]. The modules are hexagonal, a shape chosen for two independent reasons: hexagons tile a surface without gaps and approximate a sphere more closely than square modules of comparable size, and hexagonal scintillators have also been shown to offer slightly better light yield and energy resolution than square, triangular or cylindrical prisms [13]. Each module is built around a monolithic LYSO crystal [14].

The choice between segmenting the scintillator and leaving it continuous is not incidental, and it conditions how the interaction position can be recovered at all [15]. In pixelated arrays, the crystal is cut into individual elements separated by reflectors, so that the interaction is assigned to whichever element fired. Event positioning is then simple and fast, but the resolution cannot be finer than the element itself [9], and two costs follow from making the elements smaller: the reflectors occupy a growing fraction of the detector volume, reducing the packing fraction, and the number of readout channels rises. Multiplexing schemes, in which several crystals share a readout line and the position is recovered by decoding the light distribution, are commonly used to keep this cost manageable [11]. Pixelated arrays also provide no intrinsic

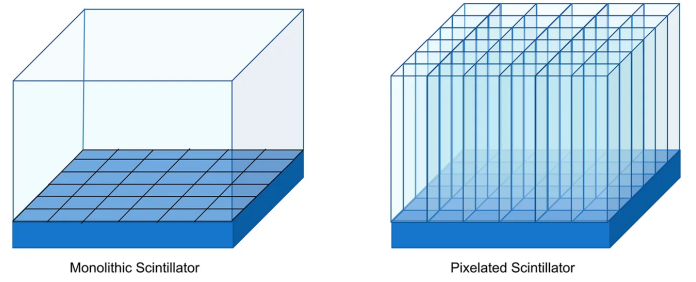


Figure 2: Monolithic (left) versus pixelated (right) scintillator configurations. In the monolithic design a continuous slab is coupled to a segmented photosensor array; in the pixelated design the crystal is cut into elements coupled one-to-one to the photosensors. Reproduced from [17] under CC BY 4.0.

depth-of-interaction (DOI) information, which has motivated layered designs such as phoswich detectors, where stacked scintillators with different decay times allow a coarse depth assignment at the price of a degraded timing resolution [11].

Monolithic scintillators take the opposite approach. A single continuous block is coupled to an array of photodetectors, and the light from each interaction spreads over many channels at once [15]. Positioning is no longer a matter of identifying an element but of estimating a coordinate from the recorded light distribution, which shifts the resolution limit from the geometry of the crystal to the quality of the positioning algorithm, and makes the DOI recoverable as a continuous variable [16]. The absence of internal reflectors also restores the packing fraction lost in segmented designs [11]. This is the configuration adopted in the scanner described above, and it is the light sharing across the array that the present work exploits.

This light distribution is read out by an array of silicon photomultipliers (SiPMs) optically coupled to the rear face of the crystal [15]. Each interaction produces signal in a subset of channels, and a readout application-specific integrated circuit (ASIC) amplifies, thresholds and digitises each of them, so that an event is fully described by the vector of charges recorded across the array. Because a channel only produces an output once it exceeds the threshold set in the electronics, most entries of that vector are zero for any given event and only a few channels contribute; the sparsity of the signal is therefore a property of the acquisition system rather than a defect of the data [18].

Estimating the interaction position from that vector admits solutions of widely varying complexity. The most common is the centre of gravity (CoG), which places the event at the average of the photodetector coordinates, each weighted by the charge that channel recorded [19]; in practice the square of the charge is normally used as the weight, which reinforces the contribution of the channels closest to the interaction point and improves resolution [20]. Its appeal is that it is deterministic, has no parameters to tune and requires no prior calibration. In exchange, it exploits only part of the information contained in the light distribution: recovering the DOI, or correcting the

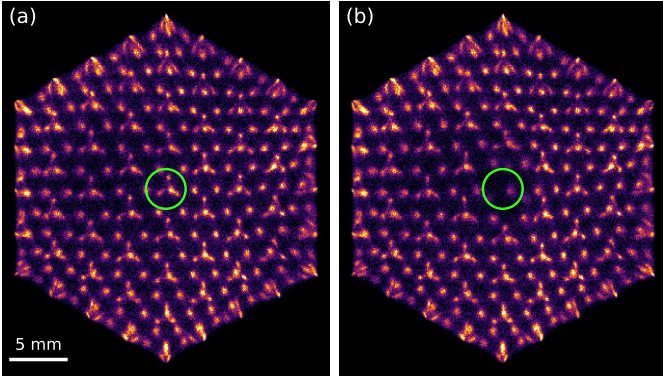


Figure 3: Flood map of the same detector module reconstructed by centre of gravity, using the same events in both panels: (a) with all 61 channels operational and (b) with one channel disabled, marked in green. The structures engraved in the crystal appear as discrete spots; the inset magnifies the neighbourhood of the disabled channel.

bias the method itself introduces near the crystal edges, calls for substantially more elaborate algorithms [21].

The two-dimensional histogram of the positions computed in this way over a large number of events forms the flood map (Fig. 7), in which the structures engraved in the crystal appear resolved as discrete spots. It is the image conventionally used to characterise a detector of this kind, and its sharpness depends on all 61 channels being operational: the CoG averages over the whole array, and a single missing term is enough to displace the centroid. The effect is confined to the neighbourhood of the failing channel — in the module of Fig. 7 the reconstructed position shifts by a median of 0.41 mm for events within 4 mm of it, and by less than 0.01 mm elsewhere — but within that neighbourhood the engraved structures are no longer resolved.

Silicon photomultipliers degrade and fail for a variety of reasons: thermal drift, manufacturing defects, deterioration of the optical coupling, or faults in the readout channel. The consequence for the image is well recognised in industrial practice: a dead pixel introduces reconstruction artefacts and incorrect activity values, and commercial systems include quality-control procedures to identify it [22]. The adopted general solution, however, is to discard the affected information and compensate for it downstream of positioning, either by duplicating events from neighbouring pixels in list mode or by interpolating in sinogram space [22]. The channel is not recovered: it is masked out.

This class of problem has motivated the application of deep learning to PET detector readout [23]. The most developed line is the estimation of the interaction position in monolithic crystals, where networks trained on calibration data consistently outperform analytical algorithms and recover the DOI as a continuous variable [24], [25]. Two previous studies address precisely this task on the present detector [18], [26]. All of them share the same framing: the light distribution recorded by

the array is taken as input and the network returns a coordinate.

Closer to what is proposed here, deep learning has also been applied to recovering channel signals rather than positions. Networks have been trained to invert multiplexed readout schemes, reconstructing the individual channel values from a reduced set of measurements and validating the result on the recovered flood map [27], [28], and the approach has recently been extended to semi-monolithic detectors, where the full 64-channel response is reconstructed from 16 row-and-column sums by exploiting the light sharing between slabs [29]. In all these cases, however, the information carried by each channel is still present in the acquired data: multiplexing encodes it in a known linear combination, and the network learns to invert a mapping fixed by design. A failed channel is a different situation, since nothing of its signal reaches the acquisition and the only remaining trace of it is the statistical correlation that light sharing induces among its neighbours.

That situation is not unique to PET. Recovering the response of defective elements in a sensor array is an established problem in imaging, addressed either by interpolating from neighbouring pixels or, more recently, by training autoencoders to reconstruct corrupted regions from their surroundings [30]. The formulation adopted here belongs to that family: it is the masked autoencoder setting [31], in which part of the input is deliberately hidden and the model is trained to reconstruct it, which allows supervision to be generated from healthy modules without requiring labelled faulty detectors. What distinguishes the present case is that the redundancy being exploited is physical rather than statistical: the light emitted in a single interaction is measured simultaneously by several photodetectors, so the value of a channel is constrained by the response of those around it.

This work proposes to recover that channel rather than discard it. If the light from each interaction spreads over several photodetectors, the charge that the failed sensor would have recorded is not entirely lost: it remains implicit in what its neighbours measured, and the problem is to estimate it. The difficulty lies less in the model than in the supervision. In a faulty module there is no way of checking whether an estimate is correct, since the true charge of the dead channel is unobservable by definition: if it could be measured, the channel would not be faulty. What cannot be observed, however, can be simulated. It suffices to take a healthy module, disable one of its channels and keep the original value as a reference; the fault is fabricated, and with it the label. The whole approach rests on that idea, and it is applied to experimental data acquired with the detector described above rather than to simulations, so that the model learns the behaviour of the system as it actually is, with its noise, its threshold and its variation between modules.

The choice of model follows from the geometry of the array. The immediate option would be to treat the vector of sixty-one charges as a list of numbers and let a dense network discover on its own which channels are related to which, but that means learning from scratch something that is already known: which sensor is adjacent to which. The photodetectors form a hexagonal tiling in which every interior sensor has six

immediate neighbours, and that neighbourhood is precisely the path along which the light is shared. A structure of this kind is naturally described as a graph, with one node per sensor and an edge between each pair of adjacent sensors, and it therefore admits a neural network that operates on it by propagating information between neighbours [32]. The model proposed here follows that approach, incorporating the geometry of the detector as prior knowledge instead of leaving it implicit in the data.

The evaluation is framed along the same lines. The error in the estimated charge is the quantity the model minimises, but on its own it says little about the usefulness of the method: what matters is how much is recovered of what the fault degrades. Three physical effects are therefore measured: the displacement of the reconstructed position, which is the direct consequence on the flood map; the conservation of the energy spectrum, which determines whether the restored signal is physically admissible; and the spatial resolution. All of this on modules the network has not seen during training, and compared against classical interpolation methods in order to establish that the complexity of the model is warranted. The behaviour of the method when more than one channel fails is also examined, with particular attention to contiguous groups, which are what is observed in real faulty detectors and also the least favourable case, since when an entire region disappears the lost sensor is left without healthy neighbours to draw on.

II. MATERIALS AND METHODS

A. Detector Module

Every measurement in this work comes from a single detector module of the scanner described above. The scintillator is a monolithic LYSO hexagonal prism, 17.6 mm on a side and 13 mm thick, laser-etched into 364 honeycomb-like optical structures that constrain how the scintillation light spreads before it reaches the photosensor [14]. The back face is optically coupled to an array of Hamamatsu silicon photomultipliers [12], each channel comprising 3980 microcells with a photon detection efficiency of approximately 35 % at the emission wavelength of LYSO [18]. The readout electronics convert the detected light into a digitised charge R_{ch} per channel [33], which is the quantity this work operates on. The lateral faces carry an absorbing black adhesive that suppresses edge reflections; as a side effect, interactions near the perimeter lose a measurable fraction of their scintillation light, which is visible later in both the per-channel charge and the energy spectrum.

Both the photosensor array and the readout electronics are prototypes developed within the group and still under active development, so we describe them functionally rather than by commercial reference.

The array provides 64 readout channels, of which 61 are instrumented in the module used here; channels 1, 16 and 18 are inactive and we excluded them throughout. Table I lists the geometry we derived directly from the physical sensor positions. The 61 active centres tile a hexagonal lattice of side 5 with a pitch of 3.75 mm, spanning $[-13.0, 13.0]$ mm

Table I: Geometry of the detector module, obtained from the physical sensor positions.

Property	Value
Crystal	Monolithic LYSO, hexagonal
Crystal side / thickness	17.6 mm / 13 mm
Etched optical structures	364
Total readout channels	64
Instrumented channels	61
Inactive channels (Ich)	1, 16, 18
Sensor pitch	3.75 mm
Extent in x	$[-13.0, 13.0]$ mm
Extent in y	$[-15.0, 15.0]$ mm
Maximum radius from centre	15.01 mm
Lattice	Hexagonal, side 5

Table II: Dataset statistics. Charges are expressed in ADC units.

Property	Value
GOOD modules	159
BAD modules	62
Mean file size	161 MB
Total volume (GOOD)	25.5 GB
Events per file	$\sim 10^6$
Events in the test partition	5 476 642
Active channels per event (mean)	17.0
Active channels per event (range)	1–54
Total charge R_T (mean)	31.9
Total charge R_T (median)	26.8

in x and $[-15.0, 15.0]$ mm in y . The ratio between these two half-extents, $13/15 = 0.867$, matches $\cos 30^\circ$ to three decimal places, confirming a regular hexagonal envelope with vertices along y . The sensor centres therefore do not reach the crystal boundary, which is consistent with the larger crystal side quoted above.

A 511 keV gamma interacting in the crystal releases on the order of 10^4 optical photons, which distribute over many channels at once. We estimated the interaction position with an Anger-type centroid weighted by the squared charge,

$$\hat{x} = \frac{\sum_c R_{ch,c}^2 x_c}{\sum_c R_{ch,c}^2}, \quad \hat{y} = \frac{\sum_c R_{ch,c}^2 y_c}{\sum_c R_{ch,c}^2}, \quad (1)$$

where (x_c, y_c) are the sensor coordinates. The two-dimensional histogram of $(\hat{x}, \hat{y})$ over many events is the flood map, and it is the image in which the 364 structures must remain separable.

B. Dataset

We used acquisitions recorded with a ^{22}Na source under flood irradiation, one binary file per physical detector module. The collection comprises 159 nominally healthy modules, referred to as GOOD, and 62 modules with at least one suspected faulty channel, referred to as BAD. Each file holds on the order of 10^6 events and occupies about 161 MB, for a total of 25.5 GB in the GOOD set. Table II summarises the statistics.

The gap between the mean and the median of R_T reflects the asymmetry of the energy distribution, with a high-energy tail

over a dominant Compton background. We applied no energy calibration and no energy window, a choice we return to in Section II-G, where the spectral metric is constructed so that this omission cancels.

We split the GOOD set *by file* rather than by event, using a fixed seed of 42: 149 modules for training, 5 for validation and 5 held out for testing (datas057, 116, 126, 202 and 214). The distinction matters. Each file is a different physical detector, so splitting by file measures generalisation to hardware the model has never seen, whereas splitting by event would measure a considerably easier problem. The test partition was reserved from the outset and used neither for training nor for model selection.

C. Problem Formulation

Let $\mathbf{r} \in \mathbb{R}^{61}$ be the vector of channel charges for one event. We modelled a faulty channel by forcing a subset $\mathcal{D} \subset \{1, \dots, 61\}$ of entries to zero, and asked the model to recover the original vector from what remains. The masked input is

$$\tilde{r}_c = \begin{cases} 0 & \text{if } c \in \mathcal{D} \\ r_c & \text{otherwise,} \end{cases} \quad (2)$$

and the network additionally receives the binary mask $m_c = \mathbb{I}[c \notin \mathcal{D}]$, so that a channel switched off is distinguishable from a channel that legitimately recorded no signal. Input and target share the same normalising factor, the maximum of the masked vector:

$$\mathbf{x} = \frac{\tilde{\mathbf{r}}}{\max_c \tilde{r}_c}, \quad \mathbf{y} = \frac{\mathbf{r}}{\max_c \tilde{r}_c}. \quad (3)$$

Because the normaliser is computed *after* masking, the target of a suppressed channel may exceed unity when that channel was the brightest of the event. The network output is therefore linear, with no sigmoid or clamp. We clipped negative charges to zero before masking, and we normalised per event rather than per dataset: for position estimation what matters is how the light distributes, not how much light there was in total.

This construction is a masked autoencoder [31] applied to detector channels instead of image patches, and it makes the task self-supervised. The ground truth is the original 61-channel vector, so no module with a genuinely dead sensor is needed for training, and the supervision requires no calibration. We generated the masks on the fly, drawing a fresh $\mathcal{D}$ for every sample, so a given event is never seen twice with the same failure pattern.

Two aspects of the sampling deserve mention. The first is class balance: half of the samples suppress a channel that did carry signal and half suppress a channel already at zero. Without this balance roughly three quarters of the examples would require no correction at all, and the model would reasonably learn to do nothing.

The second is the spatial pattern of $\mathcal{D}$, which we controlled through three regimes. In *cluster* the k suppressed channels are contiguous; in *scatter* they are drawn uniformly across the detector; and in *near* they lie within two hops of the seed without being required to touch. The motivation for the third

regime is empirical. When we labelled the BAD modules, the failures we found most often were neither compact groups nor sensors scattered across the whole array, but two or three sensors separated by one or two positions.

D. Architectures

We evaluated four families of models, all solving the same $61 \rightarrow 61$ regression.

1) *Classical baselines*: Two methods without learned parameters establish what the geometry alone provides: the mean of the neighbouring channels, and a linear regression fitted per channel on the remaining 60. Both exclude the target channel from their inputs, and we fitted the regression coefficients on training modules only.

2) *Baselines without a geometric prior*: A deep multilayer perceptron (DEEPMPLP) and a residual MLP with attention (RESIDUALMLP) take the 61 channels as a flat vector. They have access to the same information but no notion of which sensors are adjacent, which isolates the contribution of the geometric prior.

3) *HexGNN*: Our reference model treats the detector as a graph whose 61 nodes are the active sensors and whose edges join physically adjacent ones. The adjacency follows the hexagonal lattice: 312 directed edges, with 37 nodes of degree 6, 18 of degree 4 and 6 of degree 3, and full symmetry. Each layer updates a node from its own state and the mean of its neighbours,

$$h_c^{(l+1)} = \sigma \left(W_s h_c^{(l)} + W_n \frac{1}{|\mathcal{N}(c)|} \sum_{j \in \mathcal{N}(c)} h_j^{(l)} + b \right), \quad (4)$$

with σ a GELU nonlinearity. The aggregation is isotropic: all neighbours of a node share the weight matrix W_n . The reference configuration (HEXGNN-S) uses a hidden width of 48 and four blocks, for 38 305 trainable parameters. We also trained wider variants, HEXGNN-M and HEXGNN-L, to test whether capacity was the binding constraint.

4) *Architectural variants*: On top of the reference we explored nine modifications, each targeting a specific hypothesis: metric edge features (relative distances and vectors), anisotropic hexagonal convolution with one weight per direction, global attention and a graph transformer, an encoder-decoder with a latent bottleneck, multi-statistic aggregation, per-channel equalisation, heteroscedastic prediction of mean and variance, and averaging ensembles.

E. Loss Function

We computed the loss over the full 61-channel output rather than the suppressed entries alone, treating the task as ordinary regression. Our reference is the mean squared error,

$$\mathcal{L} = \frac{1}{61} \sum_{c=1}^{61} (\hat{y}_c - y_c)^2, \quad (5)$$

selected after comparing it against the mean absolute error and the Huber loss with $\delta = 0.1$. We additionally tested two physics-informed terms, one penalising the position error of

Table III: Training configuration of the reference model.

Property	Value
Modules per epoch	1
Epochs (= training modules)	149
Events per epoch	107 000
Total sample budget	15 943 000
Batch size	512
Optimiser	AdamW
Learning rate	10^{-3}
Weight decay	10^{-4}
Scheduler	Cosine, $T_{\max} = 149$
Early stopping	None
Checkpoint selection	Validation MAE, imputed channel
Validation events	400 000
Loss	MSE

Eq. 1 and one penalising the shift of the energy spectrum, both reported in Section III.

F. Training Setup

Table III lists the configuration. One aspect of the schedule requires clarification, because the word “epoch” does not carry its usual meaning here: each epoch loads *one* detector module and draws 107 000 events from it. The 149 epochs therefore correspond to the 149 training modules, each visited once, and the number is set by the partition rather than tuned. The resulting budget of 15 943 000 samples matches the one used before full coverage was adopted, so that changes in coverage are not confounded with changes in cost.

We trained with AdamW and a cosine schedule whose period equals the number of epochs. We applied no early stopping, since every epoch supplies previously unseen data, and we selected the final checkpoint by the validation error on the imputed channel, computed on a fixed set of 400 000 events with a fixed mask seed. Every run stores its preprocessing configuration inside the checkpoint, so that an evaluation cannot silently apply a normalisation different from the one used in training.

G. Evaluation Metrics

The quantity we ultimately care about is not the charge of the recovered channel but the position it yields, so our primary metric is defined on Eq. 1. For each event we computed the displacement ΔR between the true centroid and the centroid obtained after suppressing a channel, both without correction (*degraded*) and after imputation (*imputed*). The recovery is the fraction of that displacement which the model removes,

$$\text{Recovery} = 100 \times \left(1 - \frac{\Delta R_{\text{imp}}}{\Delta R_{\text{deg}}} \right), \quad (6)$$

where numerator and denominator are evaluated at the same quantile and over the same population of events. We report the value at the 90th percentile, which weights the difficult events, and we also report the mean. Averaging over the 61 channels gives the macro figures used throughout Section III, and the minimum over channels characterises the worst case.

As secondary metrics we used the mean absolute error on the imputed channel in normalised units, the bias as a

Table IV: Architecture comparison. Common campaign of 600 000 events per test file. Recovery in per cent, MAE in normalised units.

Model	Params	Rec. p90	Rec. mean	MAE
DeepMLP	500 541	51.39	45.90	0.7219
ResidualMLP	345 981	48.92	44.81	0.7299
Graphormer	4 263 881	55.17	49.65	0.6600
Graph U-Net	331 649	55.22	47.95	0.6918
PNA	165 761	58.45	52.48	0.6309
HEXGNN-S	38 305	58.02	52.66	0.6195
HEXGNN-M	225 217	58.27	52.64	0.6185
HEXGNN-L	398 593	57.47	51.96	0.6385

guardrail against systematic under-prediction, and the relative charge error. Two additional analyses address physical fidelity rather than accuracy. The first measures the point-spread function of the flood map, comparing its FWHM before and after imputation. The second measures the preservation of the energy spectrum by zones of the crystal, defined through concentric hexagonal bands, using a paired differential design: we subtract the true channel and add back the prediction over the same events, so that the absence of energy calibration cancels between the two terms.

Finally, to evaluate the pipeline on genuinely faulty hardware, we used a position-aware statistic that flags anomalous channels in the BAD modules from the robust z -score of two quantities, the fraction of events in which a channel fires and its charge ratio against its neighbours, both referenced to the baseline built from the GOOD modules. This addresses the same task that industrial practice solves by masking the affected channel and compensating downstream [22].

III. RESULTS

A. Value of the Geometric Prior

Table IV compares the architectures under a common evaluation campaign of 600 000 events per test file. The two models without a geometric prior reach 51.39 % and 48.92 % recovery at the 90th percentile, while the smallest graph model reaches 58.02 % with an order of magnitude fewer parameters. Widening the graph model, however, changes little: 58.27 % for HEXGNN-M and 57.47 % for HEXGNN-L, against 58.02 % for the 38 k-parameter reference.

To test whether the graph structure itself carries the advantage, rather than the inductive bias of message passing in general, we retrained the reference with the adjacency randomly permuted, preserving the number of nodes, the number of edges and the degree distribution. Recovery falls to 28.08 %, 27.66 % and 31.28 % across three shuffles (Fig. 4), against 58.02 % for the true adjacency.

Against the classical methods, evaluated on the same channels, the network recovers 57.9 % compared with 45.41 % for the neighbour mean and 46.56 % for the per-channel linear regression, differences of 12.49 and 11.33 percentage points respectively.

We also quantified how far the useful information extends. Fitting linear regressions restricted to concentric rings of

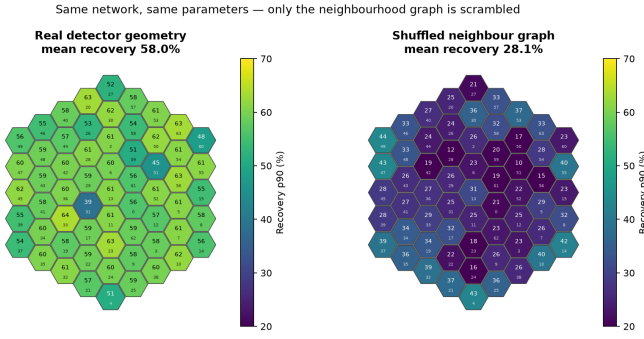


Figure 4: Effect of destroying the detector geometry. Recovery at the 90th percentile for the reference model with the true adjacency and for three independent random permutations of the graph that preserve node count, edge count and degree distribution.

Table V: Reference model and sources of variability. Values are mean $\pm$ standard deviation. n is the number of independent runs.

Source	n	Rec. p90	Rec. mean
Independent training runs	4	58.04 ± 0.37	52.15 ± 0.31
Data partition (CV)	5	56.56 ± 2.32	50.60 ± 2.04

neighbours gives 44.04 % recovery with the first ring (5.1 channels on average), 46.42 % with the second (13.7 channels) and 46.51 % with the third (24.1 channels), against 46.55 % when all 60 remaining channels are used.

B. Reference Performance and Its Variability

A single figure of merit says little without knowing how much it moves, so we measured two sources of variability separately. Table V reports both. Across four independent training runs, differing in the global seed and in the module ordering, recovery at the 90th percentile is 58.04 ± 0.37 % and the mean recovery is 52.15 ± 0.31 %.

However, repeating the split with five different partitions of the same 159 modules gives 56.56 ± 2.32 %, a standard deviation six times larger than the one observed between training runs of a fixed partition. Measuring the model on individual unseen modules makes the origin of that spread explicit: recovery varies between 29.2 % and 63.0 % depending on the module, with a standard deviation of 7.86 percentage points (Fig. 5), which places the standard error of a mean over five test modules at 3.5 points. The dominant uncertainty in this work is therefore which detectors happen to fall in the test partition, not which weights the optimiser converged to.

Performance is not uniform across the detector. Fig. 6 maps the recovery of each of the 61 channels. The worst channel of the reference band averages 42.29 ± 4.45 % over the four runs, with individual values spanning 37.54 % to 47.76 %; the channel attaining that minimum is not the same in every run, alternating between Ich 60 and Ich 31.

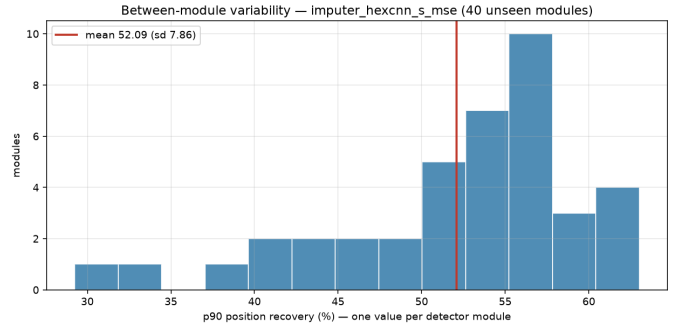


Figure 5: Recovery at the 90th percentile measured on individual unseen training modules. The spread between detectors (standard deviation of 7.86 percentage points) is an order of magnitude larger than the spread between training runs on a fixed partition.

Table VI: Recovery at the 90th percentile by failure regime, averaged over $k = 1 \dots 4$ suppressed channels. Campaign of 20 seeds and 200 000 events per file.

Training regime	Cluster	Near	Scatter
Single channel ($n=3$)	33.74 ± 11.68	38.28 ± 9.23	47.69 ± 4.95
Cluster, $k=1-4$ ($n=1$)	58.46	52.62	52.44
Near, $k=1-4$ ($n=3$)	58.31 ± 0.51	54.61 ± 0.58	53.43 ± 0.67

The effect of imputation on the reconstructed image is shown in Fig. 7, which compares the flood map of a test module with one channel suppressed against the same map after imputation.

C. Multiple Simultaneous Failures

Training with a single suppressed channel does not transfer to multiple failures. Table VI reports recovery averaged over $k = 1 \dots 4$ suppressed channels for the three spatial regimes. A model trained with $k = 1$ falls to 33.74 % on contiguous failures, and its standard deviation across runs rises to 11.68 points, since it performs adequately at $k = 1$ and collapses at $k = 4$. Models trained with k drawn from 1 to 4 stay between 53 % and 58 % in all three regimes, with standard deviations below 0.7 points.

Between the two multi-channel regimes, however, the differences are small. The model trained on the *near* regime exceeds the one trained on contiguous clusters by 1.99 points in its own regime, a difference of 2.9 standard deviations, while the two are indistinguishable on contiguous failures (-0.14 points) and on scattered ones ($+0.99$ points, 1.3 standard deviations). Fig. 8 shows the three regimes side by side and Fig. 9 the degradation as k grows.

D. Physical Fidelity

Beyond the position error, we measured whether imputation preserves the two physical quantities the detector is built to deliver.

The point-spread function of the flood map narrows from a macro-averaged FWHM of 0.1656 mm with a suppressed

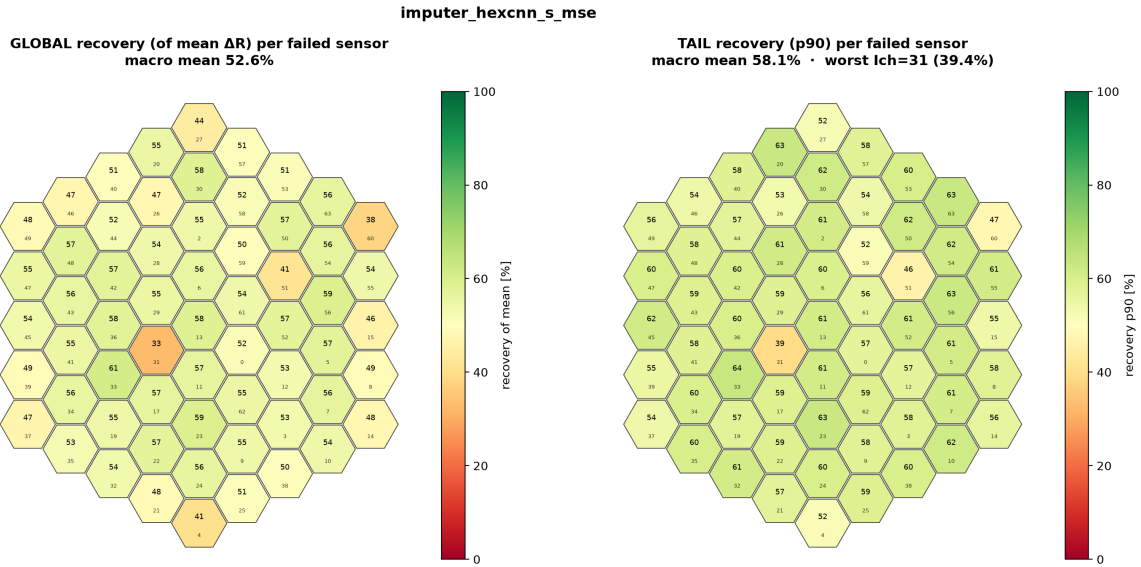


Figure 6: Per-channel recovery over the 61 active sensors. Left: recovery at the 90th percentile. Right: mean recovery. Channels at the perimeter of the hexagonal array recover less charge than central ones.

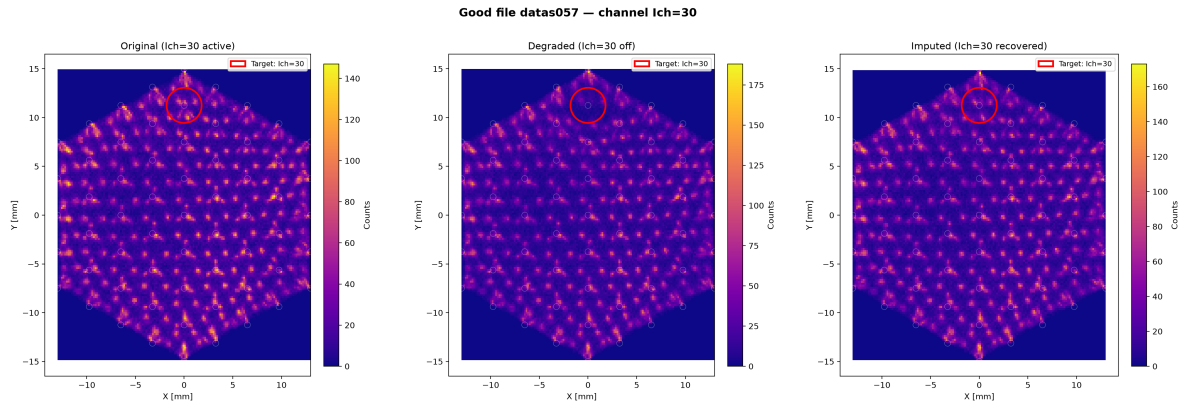


Figure 7: Flood map of a test module. From left to right: reference map with all channels active, map with one channel suppressed, and map recovered by the model.

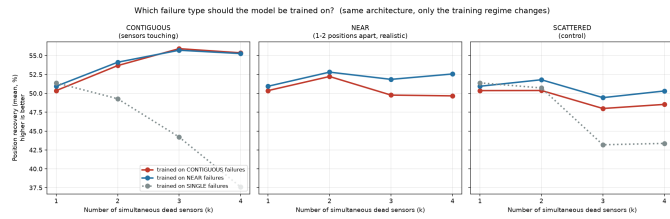


Figure 8: Recovery by failure regime for models trained under each of the three mask-generation strategies.

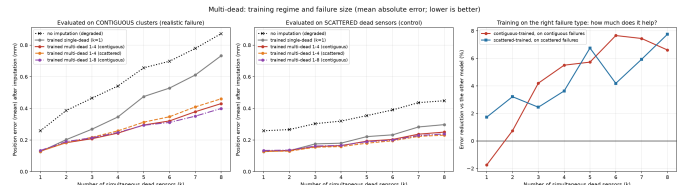


Figure 9: Recovery as a function of the number of simultaneously suppressed channels, for models trained with a single channel and with one to four.

channel to 0.0847 mm after imputation, a recovery of 48.82 % of the induced blur (Fig. 10). The positional bias falls from 0.1262 mm to 0.0218 mm, a recovery of 82.71 %.

The energy spectrum is preserved to a comparable degree and with little dependence on the region of the crystal: spectral

recovery is 83.25 % in the core, 84.32 % in the middle band and 81.19 % at the edge, for a global value of 82.92 % (Fig. 11). The residual shift of the photopeak stays between 0.20 and 0.24 ADC units across the three zones.

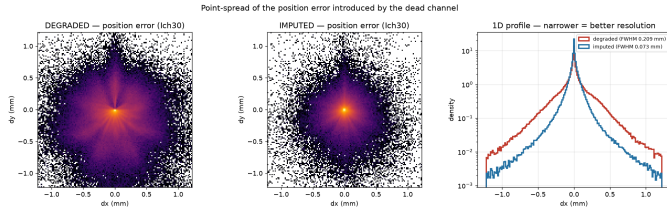


Figure 10: Point-spread function of the flood map with all channels active, with one channel suppressed, and after imputation.

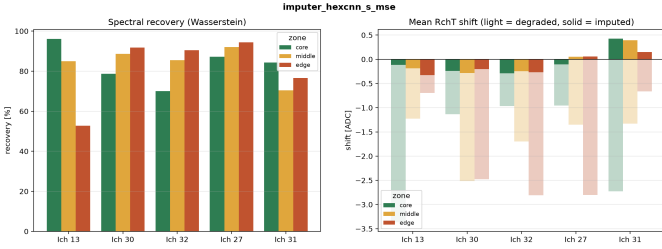


Figure 11: Energy spectrum by crystal zone, comparing the original distribution with the degraded and imputed ones. Zones are defined by concentric hexagonal bands.

E. Physics-Informed Losses and Ensembles

Adding the position error of Eq. 1 to the loss leaves recovery unchanged at 58.23 % against 58.02 % for the reference. Adding a term on the energy spectrum degrades it to 46.04 %, with the bias changing sign. Heteroscedastic prediction of mean and variance reaches 20.08 %.

Averaging the outputs of three independently trained replicas raises recovery to 58.69 %, and averaging three different architectures raises it to 58.73 %, against 58.04 ± 0.37 % for a single model.

F. Application to Faulty Modules

On the BAD set, the detection statistic identifies fully dead channels without error: the true positive rate is 1.00 for one, two, three and four simultaneously suppressed channels when the injected severity is total (Fig. 12). Under partial degradation, in which the channel retains 70 % of its response, the rate falls to 0.95 for a single channel and 0.89 for two.

We labelled a subset of the BAD modules by visual inspection of their flood maps to compare against expert criterion. At the time of writing, 18 of the 62 modules have been annotated, yielding 29 channels marked as faulty across 14 modules; the remaining 4 modules were judged to have no defective channel.

IV. DISCUSSION

Interpretacion, comparacion con el estado del arte y limitaciones.

V. CONCLUSIONS

Conclusiones y lineas futuras.

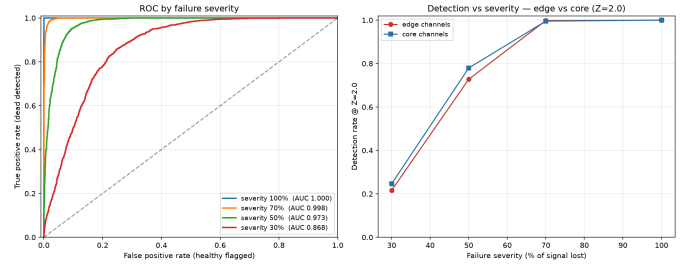


Figure 12: Detection of faulty channels by ground-truth injection on healthy modules. ROC curves for different numbers of simultaneously affected channels.

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